

Can Bach's Music Be Modeled as an Ecosystem?

A Computational Study of Pitch-Class Competition in Six Bach Works

Why does tonal music sound “natural”? Music theory tells us which intervals are consonant, but it does not explain why certain pitches become stable centers. This project asks: if the twelve pitch classes are treated as competing species in an ecosystem, do they converge to the tonic triad?

Six Bach works were analyzed. Each note was reduced to its pitch class. A 12x12 competition matrix was defined: consonant intervals promote growth, dissonant intervals suppress it. A 1000-generation simulation was run for each piece.

Four of six works converged to the tonic triad. Two works showed a different pattern: the tonic was displaced by the leading tone (BWV 784, BWV 565).

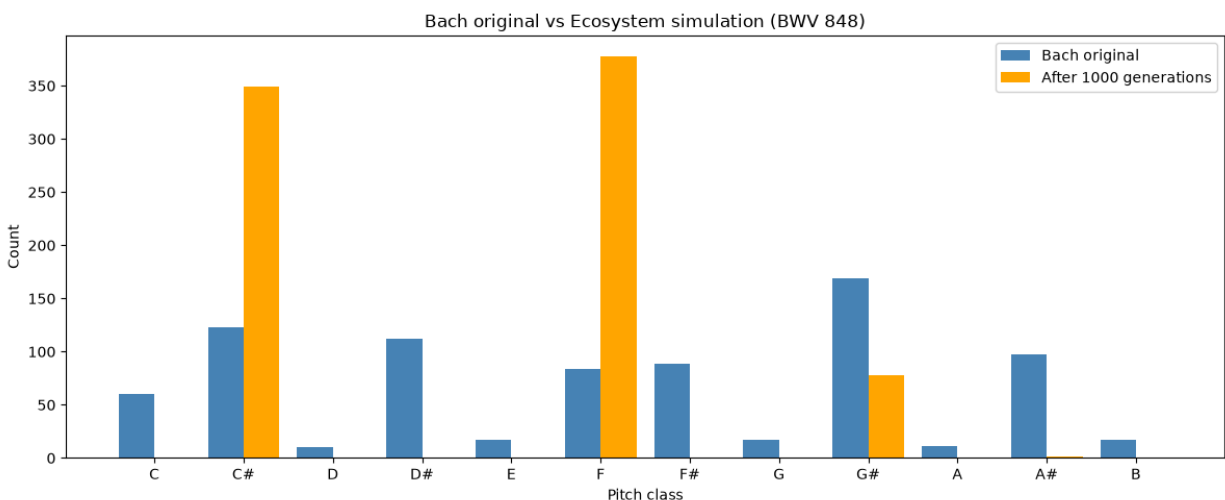


Figure 1. Bach original (blue) vs. simulation after 1000 generations (orange) for BWV 848 (C-Sharp Major). The simulation converges to the tonic triad C#--F--G#.

The ecosystem model captures a real feature of Bach's music: the tonic triad is the natural stable state of consonant pitch competition. But the exceptions reveal a boundary condition. Bach's music is anchored in ecological stability but not fully determined by it.

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